

FOR OFFICIAL USE



National  
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**X857/75/01**

**Physics**  
**Section 1 — Answer grid**  
**and Section 2**

THURSDAY, 15 MAY

1:00 PM – 3:30 PM



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Fill in these boxes and read what is printed below.

Full name of centre

Town

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Date of birth

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**Total marks — 135**

**SECTION 1 — 25 marks**

Attempt ALL questions.

Instructions for completion of Section 1 are given on *page 02*.

**SECTION 2 — 110 marks**

Attempt ALL questions.

Reference may be made to the data sheet on *page 02* of the question paper X857/75/02 and to the relationships sheet X857/75/11.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. Score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.



\* X 8 5 7 7 5 0 1 0 1 \*

## DATA SHEET

### *Speed of light in materials*

| Material       | Speed in $\text{m s}^{-1}$ |
|----------------|----------------------------|
| Air            | $3.0 \times 10^8$          |
| Carbon dioxide | $3.0 \times 10^8$          |
| Diamond        | $1.2 \times 10^8$          |
| Glass          | $2.0 \times 10^8$          |
| Glycerol       | $2.1 \times 10^8$          |
| Water          | $2.3 \times 10^8$          |

### *Speed of sound in materials*

| Material       | Speed in $\text{m s}^{-1}$ |
|----------------|----------------------------|
| Aluminium      | 5200                       |
| Air            | 340                        |
| Bone           | 4100                       |
| Carbon dioxide | 270                        |
| Glycerol       | 1900                       |
| Muscle         | 1600                       |
| Steel          | 5200                       |
| Tissue         | 1500                       |
| Water          | 1500                       |

### *Gravitational field strengths*

|         | Gravitational field strength on the surface in $\text{N kg}^{-1}$ |
|---------|-------------------------------------------------------------------|
| Earth   | 9.8                                                               |
| Jupiter | 23                                                                |
| Mars    | 3.7                                                               |
| Mercury | 3.7                                                               |
| Moon    | 1.6                                                               |
| Neptune | 11                                                                |
| Saturn  | 9.0                                                               |
| Sun     | 270                                                               |
| Uranus  | 8.7                                                               |
| Venus   | 8.9                                                               |

### *Specific heat capacity of materials*

| Material  | Specific heat capacity in $\text{J kg}^{-1} \text{ } ^\circ\text{C}^{-1}$ |
|-----------|---------------------------------------------------------------------------|
| Alcohol   | 2350                                                                      |
| Aluminium | 902                                                                       |
| Copper    | 386                                                                       |
| Glass     | 500                                                                       |
| Ice       | 2100                                                                      |
| Iron      | 480                                                                       |
| Lead      | 128                                                                       |
| Oil       | 2130                                                                      |
| Water     | 4180                                                                      |

### *Specific latent heat of fusion of materials*

| Material       | Specific latent heat of fusion in $\text{J kg}^{-1}$ |
|----------------|------------------------------------------------------|
| Alcohol        | $0.99 \times 10^5$                                   |
| Aluminium      | $3.95 \times 10^5$                                   |
| Carbon dioxide | $1.80 \times 10^5$                                   |
| Copper         | $2.05 \times 10^5$                                   |
| Iron           | $2.67 \times 10^5$                                   |
| Lead           | $0.25 \times 10^5$                                   |
| Water          | $3.34 \times 10^5$                                   |

### *Melting and boiling points of materials*

| Material  | Melting point in $^\circ\text{C}$ | Boiling point in $^\circ\text{C}$ |
|-----------|-----------------------------------|-----------------------------------|
| Alcohol   | -98                               | 65                                |
| Aluminium | 660                               | 2470                              |
| Copper    | 1077                              | 2567                              |
| Lead      | 328                               | 1737                              |
| Iron      | 1537                              | 2737                              |
| Water     | —                                 | 100                               |

### *Specific latent heat of vaporisation of materials*

| Material       | Specific latent heat of vaporisation in $\text{J kg}^{-1}$ |
|----------------|------------------------------------------------------------|
| Alcohol        | $11.2 \times 10^5$                                         |
| Carbon dioxide | $3.77 \times 10^5$                                         |
| Glycerol       | $8.30 \times 10^5$                                         |
| Turpentine     | $2.90 \times 10^5$                                         |
| Water          | $22.6 \times 10^5$                                         |

### *Radiation weighting factors*

| Type of radiation | Radiation weighting factor |
|-------------------|----------------------------|
| alpha             | 20                         |
| beta              | 1                          |
| fast neutrons     | 10                         |
| gamma             | 1                          |
| slow neutrons     | 3                          |
| X-rays            | 1                          |

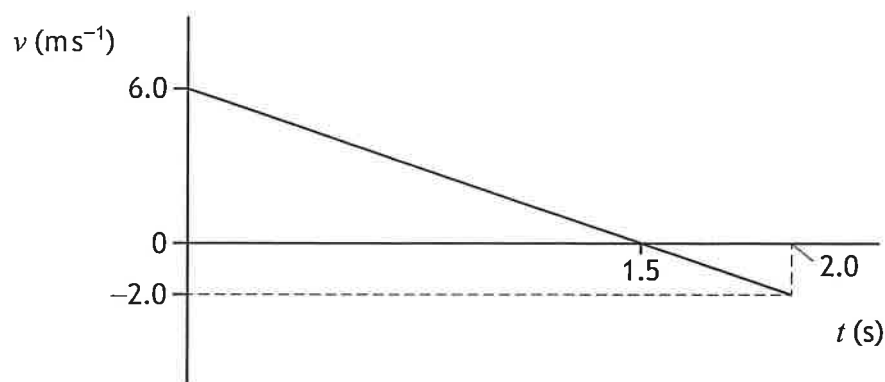
SECTION 1 — 25 marks

Attempt ALL questions

1. Which row in the table shows both quantities classified correctly?

|   | Scalar   | Vector   |
|---|----------|----------|
| A | weight   | force    |
| B | distance | velocity |
| C | mass     | distance |
| D | force    | mass     |
| E | velocity | time     |

2. The graph shows how the velocity  $v$  of an object varies with time  $t$ .



The displacement of the object at 2.0 seconds is

- A 4.0 m
- B 5.0 m
- C 6.0 m
- D 8.0 m
- E 10.0 m.

[Turn over

3. A car is travelling along a straight level road.

The car slows down at a constant rate of  $2.0 \text{ m s}^{-2}$  for a time of  $8.0 \text{ s}$ .

The final speed of the car is  $5.0 \text{ m s}^{-1}$ .

The initial speed of the car is

- A  $7.0 \text{ m s}^{-1}$
- B  $11 \text{ m s}^{-1}$
- C  $16 \text{ m s}^{-1}$
- D  $21 \text{ m s}^{-1}$
- E  $80 \text{ m s}^{-1}$ .

4. A cup rests on a table that rests on the Earth. The cup exerts a downward force on the table.



Which of the following is the reaction to this force?

- A The force of the table on the cup
  - B The force of the cup on the Earth
  - C The force of the Earth on the cup
  - D The force of the cup on the table
  - E The force of the table on the Earth
5. A crate of mass  $180 \text{ kg}$  is lifted through a height of  $3.0 \text{ m}$  in  $2.5 \text{ s}$ .
- The minimum work done in lifting the crate through this height is
- A  $60 \text{ J}$
  - B  $540 \text{ J}$
  - C  $1800 \text{ J}$
  - D  $2200 \text{ J}$
  - E  $5300 \text{ J}$ .

6. An asteroid of mass 25 kg is travelling at a speed of  $9200 \text{ m s}^{-1}$ .  
On entering the Earth's atmosphere, the asteroid burns up completely.  
The maximum amount of heat energy released is

- A  $1.2 \times 10^5 \text{ J}$
- B  $2.3 \times 10^5 \text{ J}$
- C  $2.9 \times 10^6 \text{ J}$
- D  $1.1 \times 10^9 \text{ J}$
- E  $2.1 \times 10^9 \text{ J}$ .

7. The speed  $v$  of a satellite in orbit around the Earth is given by the relationship

$$v = \sqrt{rg}$$

where:  $r$  is the total distance from the centre of the Earth to the satellite in m  
 $g$  is the gravitational field strength at the location of the satellite in  $\text{N kg}^{-1}$ .

The radius of the Earth is  $6.4 \times 10^6 \text{ m}$ .

The height of a satellite above the surface of the Earth is  $3.6 \times 10^7 \text{ m}$ .

The gravitational field strength at this height is  $0.22 \text{ N kg}^{-1}$ .

The speed of the satellite is

- A  $1200 \text{ m s}^{-1}$
- B  $2800 \text{ m s}^{-1}$
- C  $3100 \text{ m s}^{-1}$
- D  $7900 \text{ m s}^{-1}$
- E  $20\,000 \text{ m s}^{-1}$ .

[Turn over

8. The weight of an astronaut on Earth is 710 N.

The weight of the astronaut on Mars is

- A 72 N
- B 190 N
- C 270 N
- D 710 N
- E 2600 N.

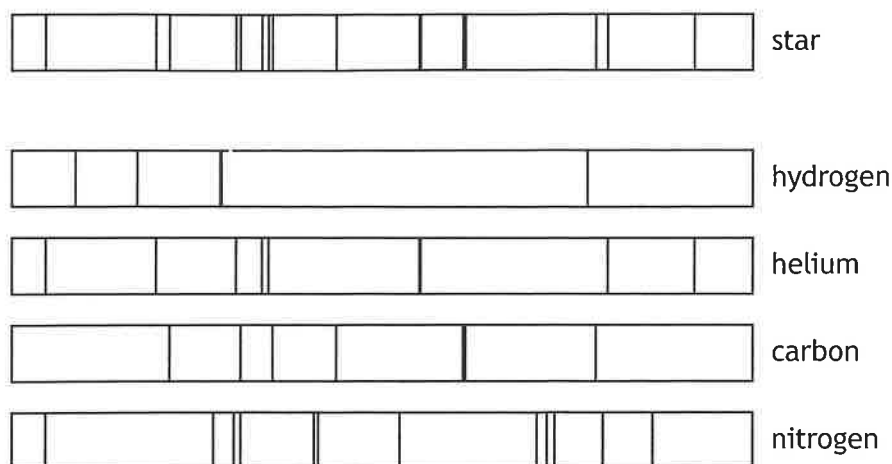
9. Sirius is the brightest star in the night sky.

The distance from Earth to Sirius is 8.6 light-years.

This distance is equivalent to

- A  $2.7 \times 10^8$  m
- B  $2.3 \times 10^{13}$  m
- C  $1.4 \times 10^{15}$  m
- D  $9.5 \times 10^{15}$  m
- E  $8.1 \times 10^{16}$  m.

10. The line spectrum from a star is shown, along with the line spectra of the elements hydrogen, helium, carbon, and nitrogen.

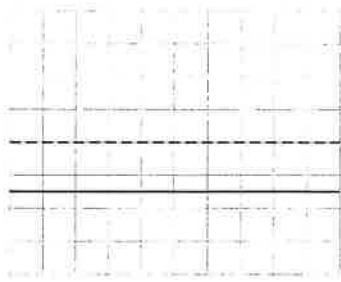


The elements present in this star are

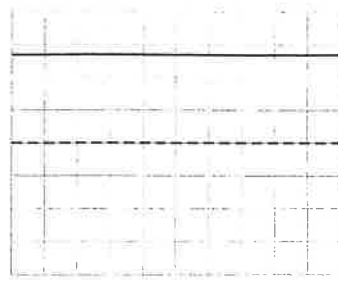
- A hydrogen and helium
- B hydrogen and carbon
- C helium and carbon
- D helium and nitrogen
- E carbon and nitrogen.

[Turn over

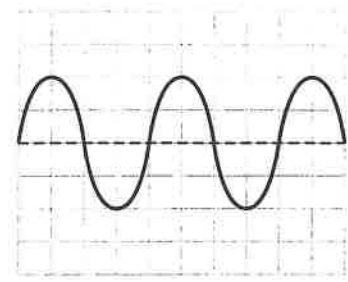
11. A technician uses an oscilloscope to test three different power supplies.  
The diagrams represent the traces seen on the screen of the oscilloscope.



trace X



trace Y



trace Z

Which row in the table identifies trace X, trace Y, and trace Z?

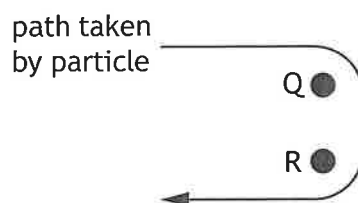
|   | trace X | trace Y | trace Z |
|---|---------|---------|---------|
| A | d.c.    | d.c.    | d.c.    |
| B | a.c.    | d.c.    | d.c.    |
| C | a.c.    | a.c.    | d.c.    |
| D | a.c.    | a.c.    | a.c.    |
| E | d.c.    | d.c.    | a.c.    |



12. An electric field exists around two point charges Q and R.

The diagram shows the path taken by a charged particle as it travels through the field.

The motion of the particle is as shown.

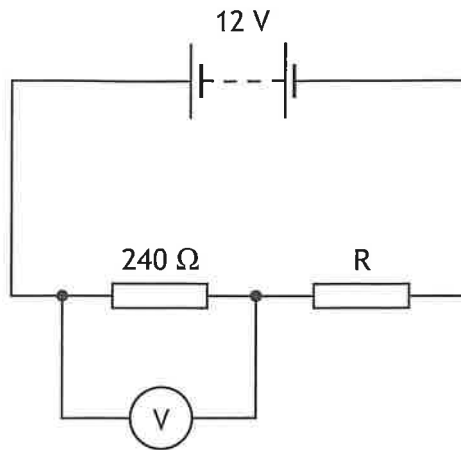


Which row in the table identifies the charge on the particle, the charge on Q, and the charge on R?

|   | Charge on particle | Charge on Q | Charge on R |
|---|--------------------|-------------|-------------|
| A | positive           | negative    | negative    |
| B | negative           | negative    | positive    |
| C | negative           | positive    | negative    |
| D | positive           | negative    | positive    |
| E | positive           | positive    | positive    |

[Turn over

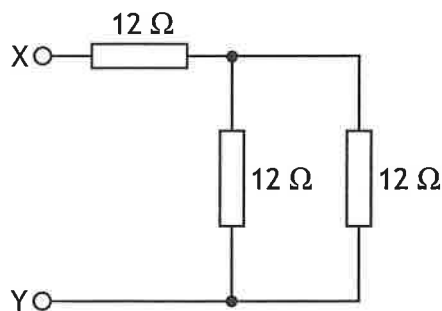
13. A circuit is set up as shown.



The reading on the voltmeter is 3.0 V.

The resistance of resistor R is

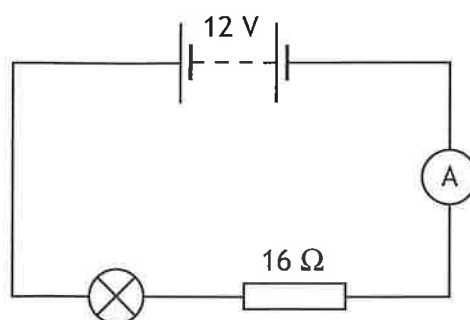
- A 20  $\Omega$
  - B 80  $\Omega$
  - C 320  $\Omega$
  - D 720  $\Omega$
  - E 960  $\Omega$ .
14. Three resistors are connected as shown.



The total resistance between X and Y is

- A 4  $\Omega$
- B 6  $\Omega$
- C 8  $\Omega$
- D 18  $\Omega$
- E 36  $\Omega$ .

15. A circuit is set up as shown.



The reading on the ammeter is 0.50 A.

The resistance of the lamp is  $8.0\ \Omega$ .

The power dissipated in the lamp is

- A 2.0 W
- B 4.0 W
- C 6.0 W
- D 9.0 W
- E 18 W.

[Turn over

16. A kettle, a television, and a microwave are connected to the mains supply.

The kettle has a power rating of 2.6 kW.

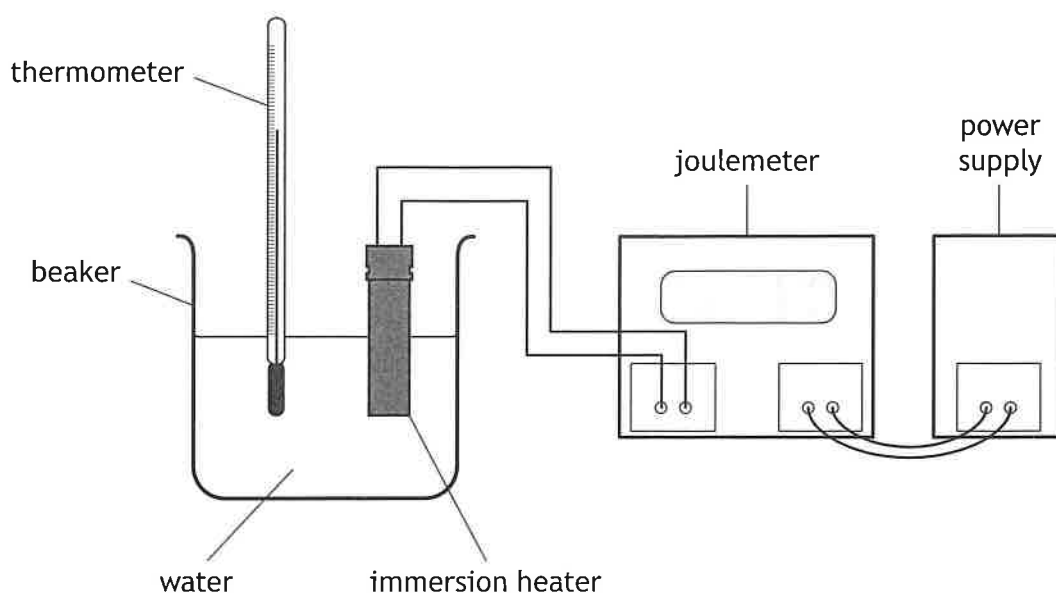
The television has a power rating of 170 W.

The microwave has a power rating of 900 W.

Which row in the table shows the appropriate fuse rating for the kettle, the television, and the microwave?

|   | Appropriate<br>fuse rating for<br>kettle (A) | Appropriate<br>fuse rating for<br>television (A) | Appropriate<br>fuse rating for<br>microwave (A) |
|---|----------------------------------------------|--------------------------------------------------|-------------------------------------------------|
| A | 3                                            | 13                                               | 3                                               |
| B | 13                                           | 3                                                | 13                                              |
| C | 13                                           | 13                                               | 3                                               |
| D | 3                                            | 3                                                | 13                                              |
| E | 13                                           | 3                                                | 3                                               |

17. A group of students sets up the experiment shown to determine a value for the specific heat capacity of water.



The students make the following statements about how the experimental procedure could be improved.

- I Insulate the beaker to reduce heat lost to the surroundings.
- II Move the immersion heater further into the water.
- III Use a stopwatch to measure the time for which the water is heated.

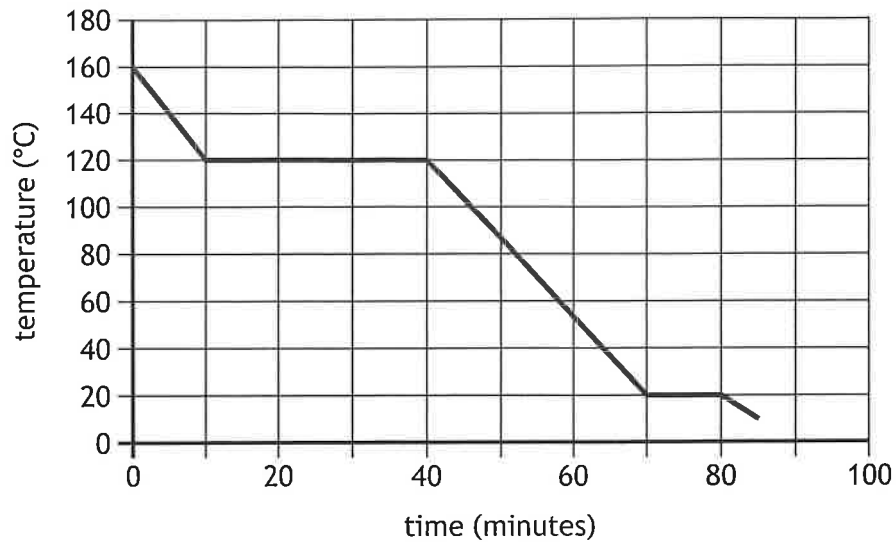
Which of these statements is/are correct?

- A I only
- B II only
- C I and II only
- D I and III only
- E I, II and III

[Turn over

18. A hot substance is cooled.

The graph shows how the temperature of the substance changes with time.



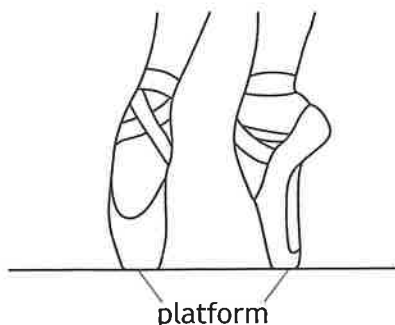
A student makes the following statements about the substance.

- I The boiling point of the substance is 120 °C.
- II At a time of 40 minutes, the substance is a gas.
- III The melting point of the substance is 20 °C.

Which of these statements is/are correct?

- A I only
- B II only
- C I and III only
- D II and III only
- E I, II and III

19. A ballet dancer wears shoes that have a flat section on the front called a platform.  
When the ballet dancer stands on the tips of their toes, the platform on each shoe is flat on the floor.

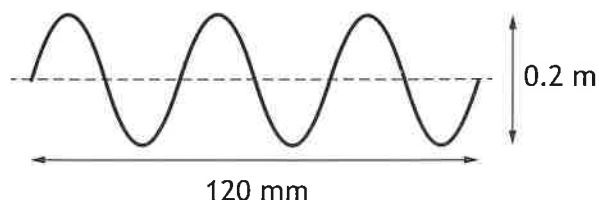


The area of the platform on each shoe is  $1.6 \times 10^{-3} \text{ m}^2$ .  
The weight of the ballet dancer is 520 N.  
The pressure exerted on the floor by the ballet dancer is

- A  $3.1 \times 10^{-6} \text{ Pa}$
  - B  $6.2 \times 10^{-6} \text{ Pa}$
  - C  $8.3 \times 10^{-1} \text{ Pa}$
  - D  $1.6 \times 10^5 \text{ Pa}$
  - E  $3.3 \times 10^5 \text{ Pa}$ .
20. A tennis ball contains a fixed mass of air at a temperature of  $20.0^\circ\text{C}$ .  
The pressure of the air inside the ball is  $1.90 \times 10^5 \text{ Pa}$ .  
During a tennis match the temperature of the air inside the ball rises to  $35.0^\circ\text{C}$ .  
The volume of the air remains constant.  
The pressure of the air inside the ball is now
- A  $1.09 \times 10^5 \text{ Pa}$
  - B  $1.81 \times 10^5 \text{ Pa}$
  - C  $1.87 \times 10^5 \text{ Pa}$
  - D  $2.00 \times 10^5 \text{ Pa}$
  - E  $3.33 \times 10^5 \text{ Pa}$ .

[Turn over

21. The diagram represents a wave.



Which row in the table shows the wavelength and amplitude of the wave?

|   | Wavelength (mm) | Amplitude (m) |
|---|-----------------|---------------|
| A | 20              | 0.1           |
| B | 40              | 0.1           |
| C | 40              | 0.2           |
| D | 120             | 0.1           |
| E | 120             | 0.2           |

22. A teacher gives a group of students two clues to identify a particular band of the electromagnetic spectrum.

*Clue 1: Diffracts around obstacles more than visible light.*

*Clue 2: Has a greater frequency than microwave radiation.*

The band of the electromagnetic spectrum these clues refer to is

- A gamma rays
- B infrared
- C ultraviolet
- D radio
- E X-rays.



23. A radioactive source has an average activity of 5.2 MBq over a time of 1.2 hours.  
The number of nuclear disintegrations that occur in the source in this time is
- A  $1.2 \times 10^3$
  - B  $4.3 \times 10^6$
  - C  $6.2 \times 10^6$
  - D  $3.7 \times 10^8$
  - E  $2.2 \times 10^{10}$ .
24. A sample of tissue is exposed to a source of alpha particles.  
The equivalent dose received from this source during a time of 5 minutes is 480  $\mu\text{Sv}$ .  
The equivalent dose rate is
- A  $1.6 \mu\text{Sv s}^{-1}$
  - B  $96 \mu\text{Sv s}^{-1}$
  - C  $2400 \mu\text{Sv s}^{-1}$
  - D  $9600 \mu\text{Sv s}^{-1}$
  - E  $144\,000 \mu\text{Sv s}^{-1}$ .
25. Energy is released when two small nuclei combine during a nuclear reaction to form a larger nucleus.  
This is a description of
- A nuclear fission
  - B nuclear fusion
  - C alpha decay
  - D beta decay
  - E gamma emission.

[END OF SECTION 1. NOW ATTEMPT THE QUESTIONS IN SECTION 2 OF  
YOUR QUESTION AND ANSWER BOOKLET]